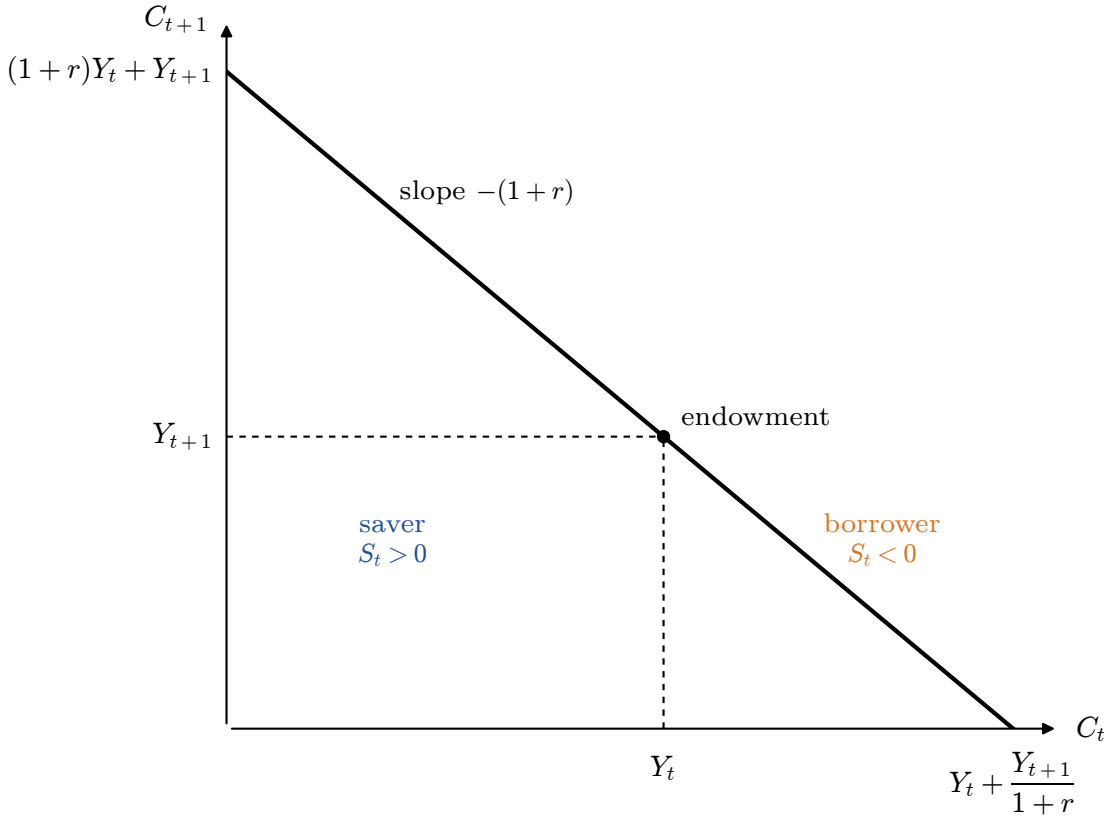


Fig. 1: The intertemporal budget constraint always passes through the endowment point.

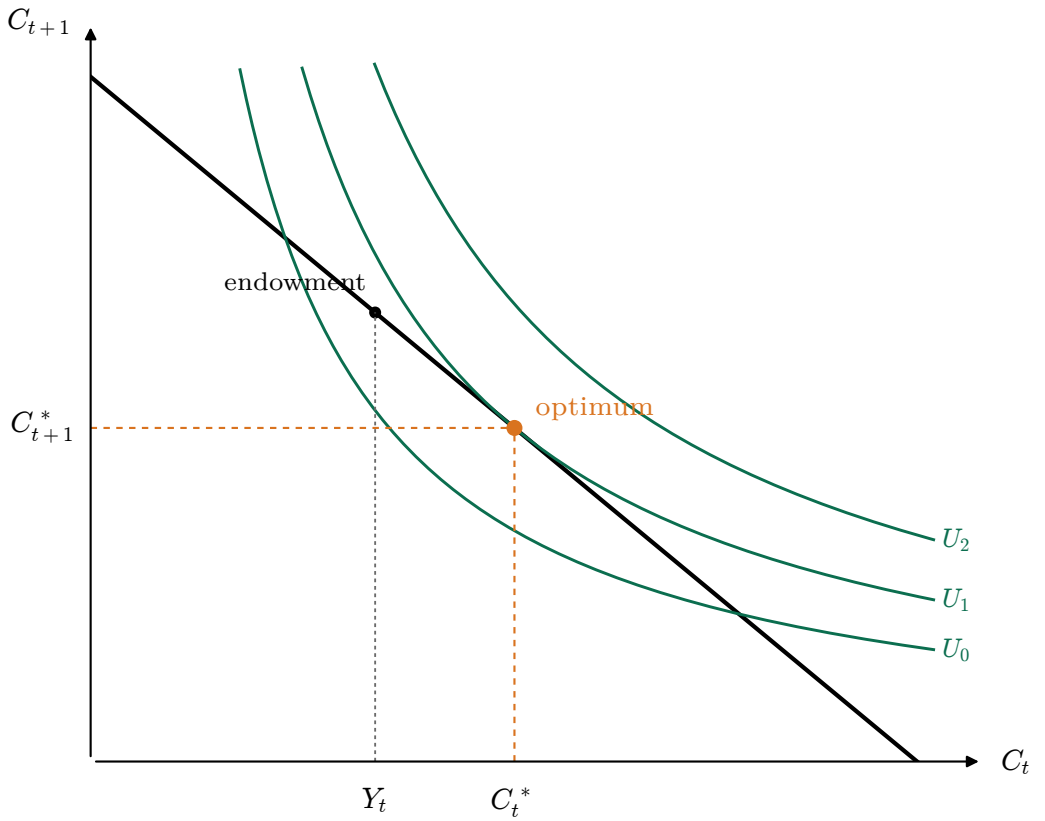


Fig. 2: The optimum is the tangency point: $\dfrac{u'(C_t)}{\beta u'(C_{t+1})} = 1 + r$ (Euler equation). Here $C_t^* > Y_t$: the household borrows.

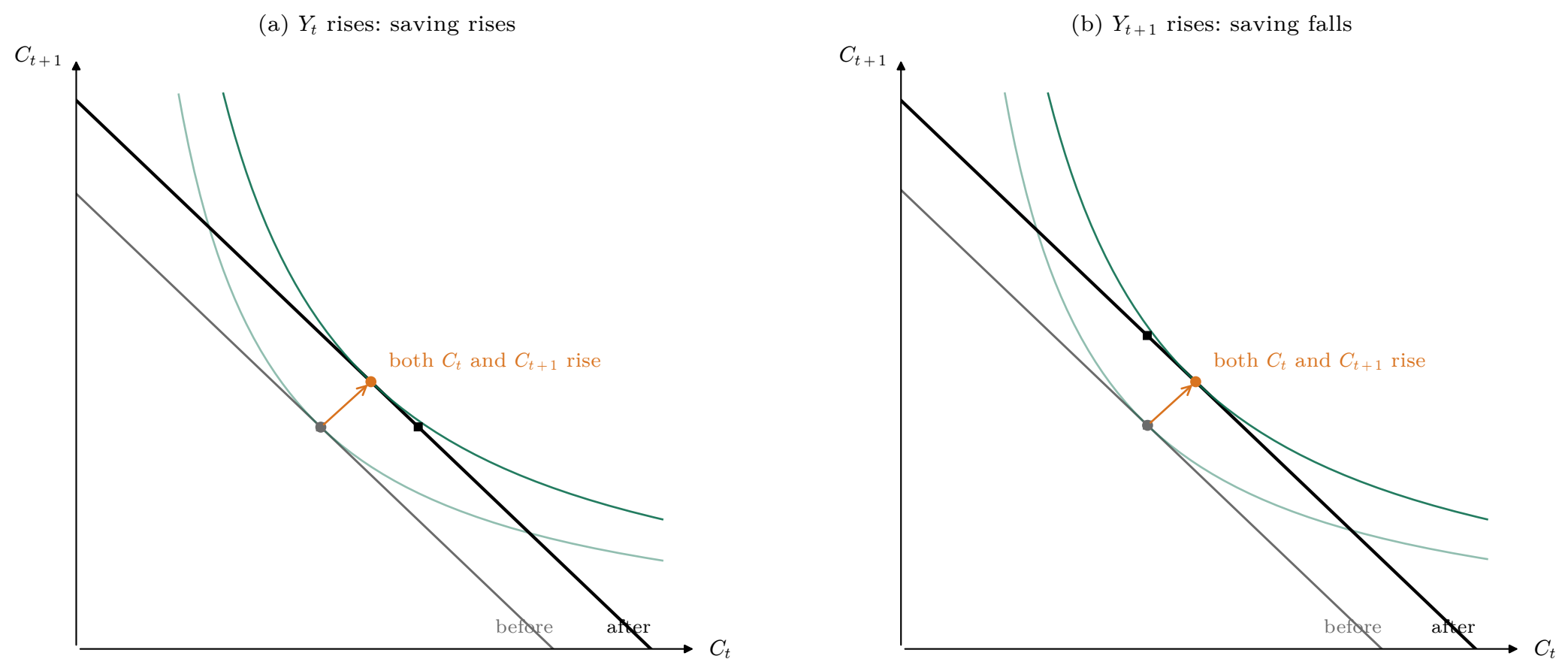


Fig. 3: Whenever income rises, the household raises consumption in both periods: it smooths consumption by adjusting saving.

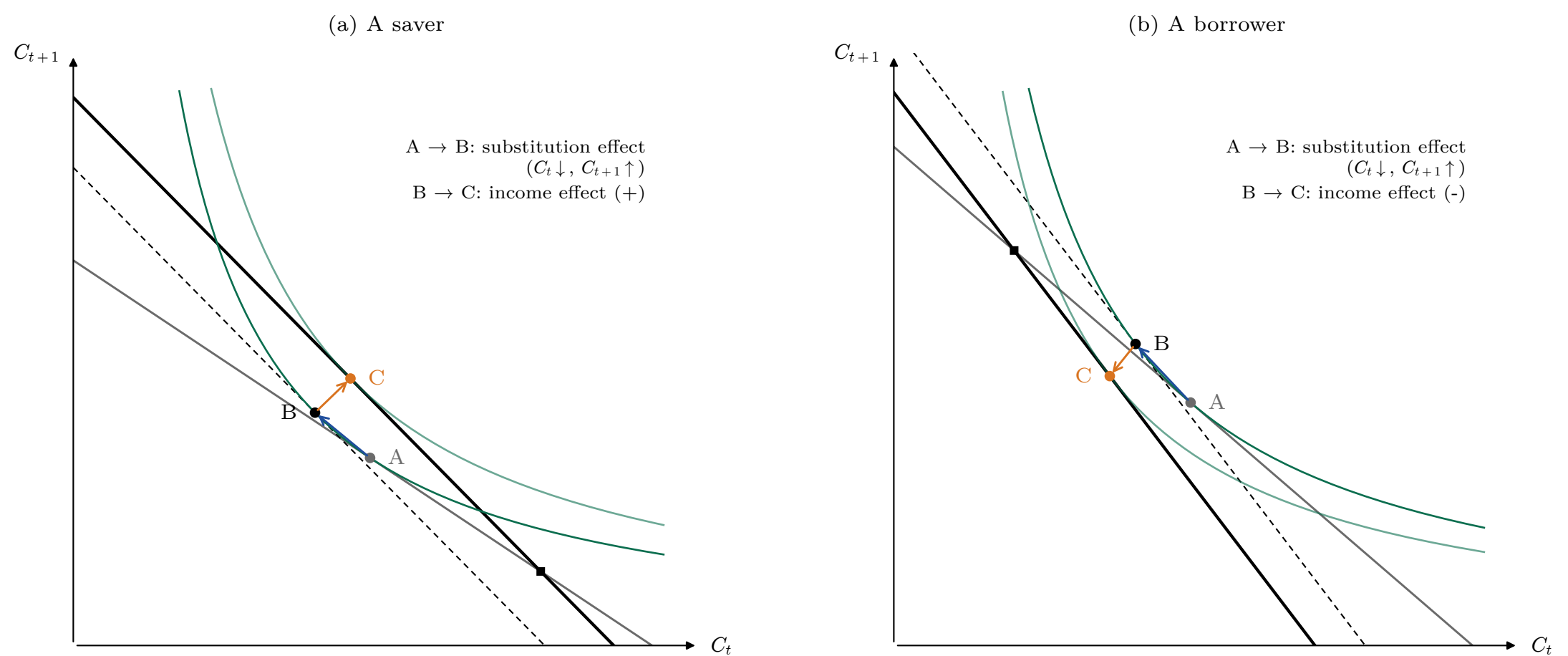


Fig. 4: The interest rate rises from 5% to 60% (exaggerated for visibility): the budget line pivots around the endowment. The substitution effect always lowers C_t; the income effect helps savers and hurts borrowers.

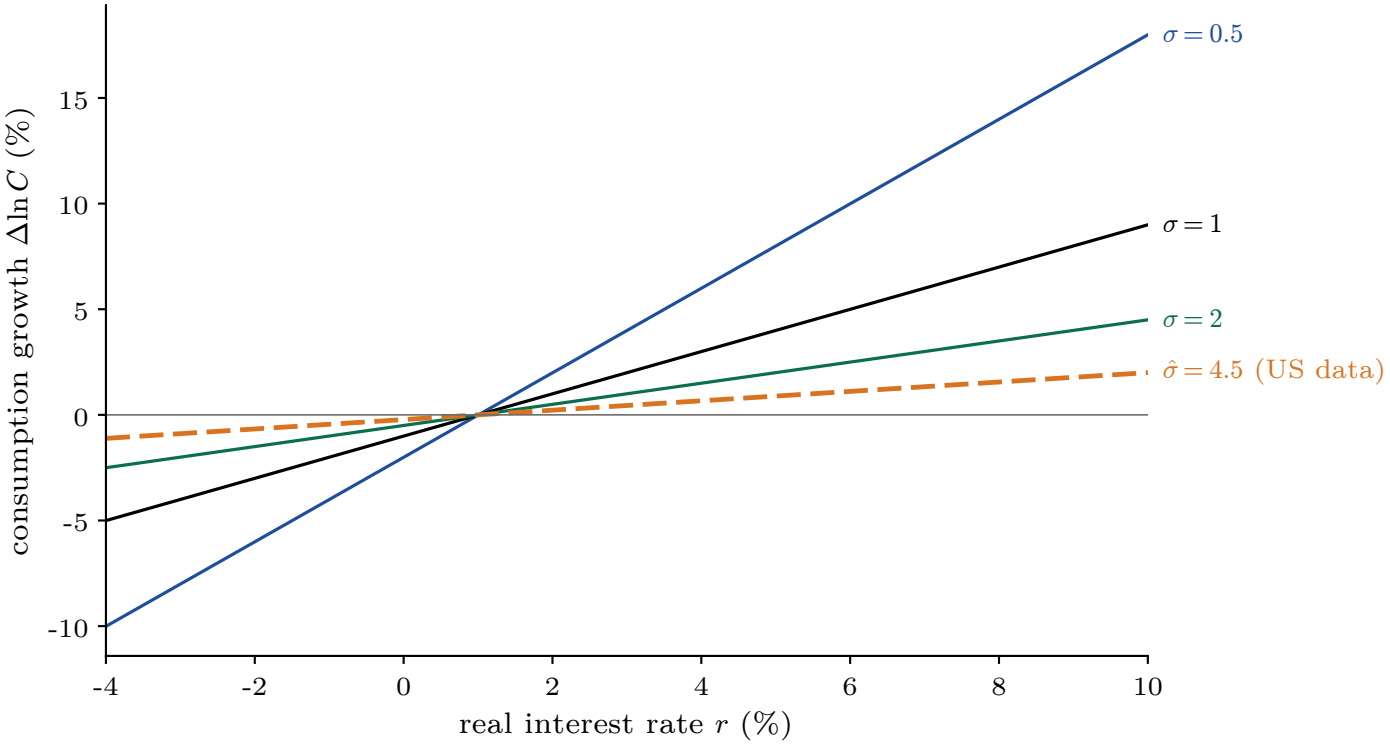


Fig. 5: CRRA Euler equation: $\Delta \ln C_{t+1} \approx \frac{1}{\sigma}(\ln \beta + r)$. The higher σ, the less consumption growth reacts to r.

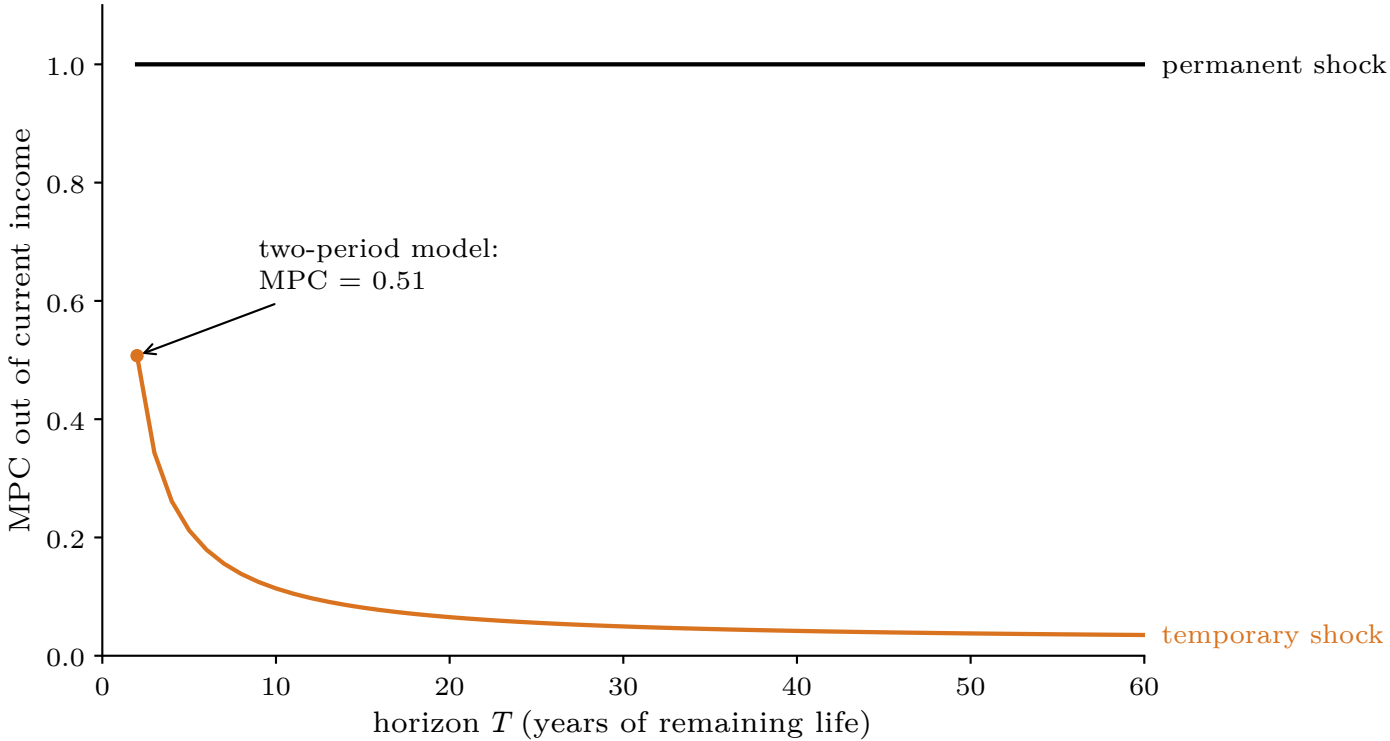


Fig. 6: With $\beta(1+r)=1$ and $r = 3\%$, a temporary windfall is spread over the whole horizon: the longer the horizon, the lower the MPC. A permanent rise in income is consumed one for one.

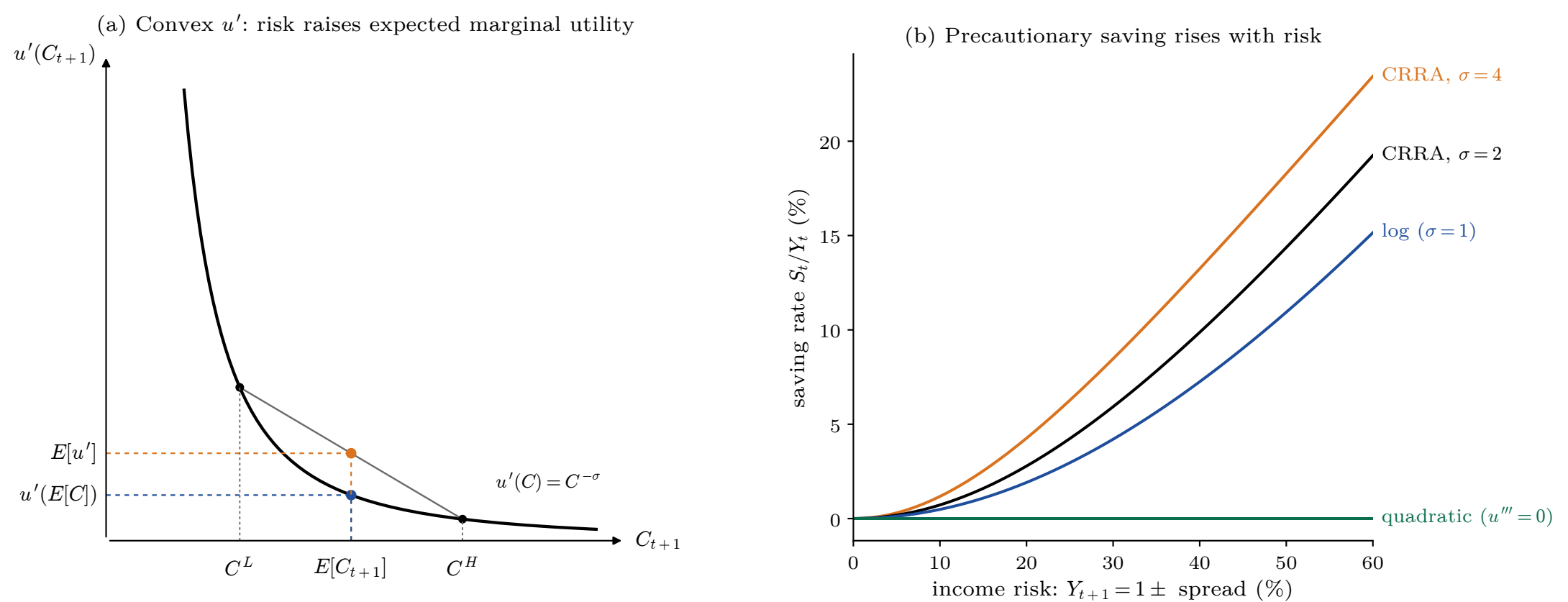


Fig. 7: With prudent preferences ($u''' > 0$), uncertainty about future income raises saving today. With quadratic utility, only expected income matters and there is no precautionary saving ($\beta(1+r) = 1$, $Y_t = E[Y_{t+1}] = 1$).

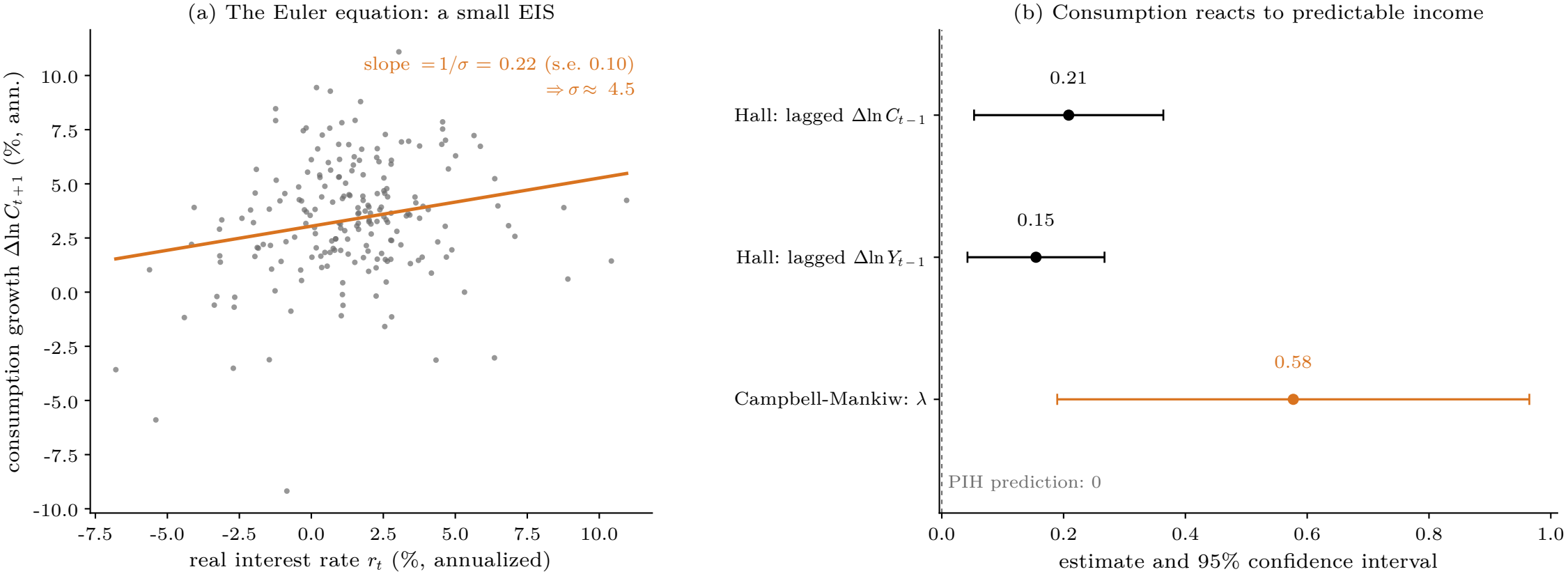


Fig. 8: US quarterly data, 1959-2009. Consumption growth barely reacts to the interest rate, and it is predictable from past income, against the pure permanent income hypothesis. The Campbell-Mankiw estimate suggests about half of 'rule-of-thumb' consumers, but its instruments are weak (first-stage F = 1.9): it should be read with caution.

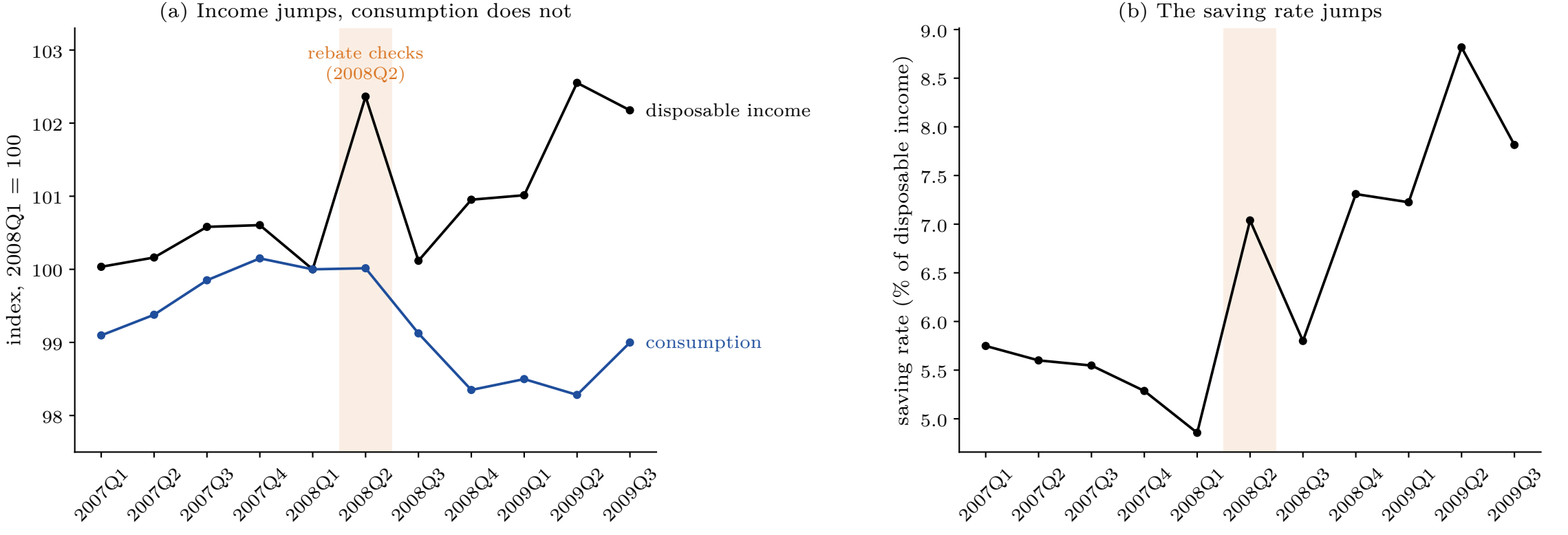


Fig. 9: In 2008Q2, real disposable income rose by \$232bn (annual rate) while consumption rose by \$1.4bn: in the aggregate, the immediate MPC out of this temporary windfall looks close to 1%, in line with Fig. 6. But the quarter is confounded by the oil shock and the onset of the recession; household-level studies exploiting the random timing of the checks find much larger spending responses.